3-1 Threats and Countermeasures in Information Security [1]

Point!

1 Information Security

(1) ([1]**Information security**) : The act of properly managing information and keeping it safe.

(2) The three essential elements of information security

[1] ([2]**Confidentiality**) : A state in which only authorized individuals can access the information.

[2] ([3]**Integrity**) : A state in which the information is not destroyed, tampered with, or erased.

[3] ([4]**Availability**) : A state in which information can be accessed at any time when needed.

2 Various threats to information security

(1) ([5]**Unauthorized access**) : When someone without permission gains access to a computer system.

(2) ([6]**Cracking**) : The act of unlawfully accessing a system to tamper with, erase, or steal data. A person who commits such acts is called a ([7]**cracker**).

(3) ([8]**Malware**) : A general term for malicious programs designed to harm computers. Infection can occur through websites, email attachments, USB drives, or networks.

[1] ([9]**Computer virus**) : A program designed to intentionally cause harm, such as destroying data or programs.

[2] ([10]**Trojan horse**) : A program disguised as a legitimate one that infiltrates a system and quietly initiates attacks.

[3] ([11]**Worm**) : A program that replicates itself and spreads across the internet like a worm, expanding the infection.

[4] ([12]**Spyware**) : A program that collects personal information without the user's knowledge and sends it to third parties.

- ([13]**Keylogger**) : Software that monitors and records keystrokes.
- ([14]**Adware**) : A program that displays unwanted advertisements without the user's consent.

[5] ([15]**Ransomware**) : A program that renders data inaccessible and demands a ransom to restore access.

(4) ([16]**Cybercrime**) : Criminal acts committed over computer networks.

[1] ([17]**Violation of the Unauthorized Computer Access Law**) : Illegally accessing a computer using someone else's user ID or password.

[2] Crimes involving computer or electronic records : Crimes involving tampering with stored data or unauthorized manipulation of devices.

[3] Network-based crimes : Crimes committed using networks, such as fraud, defamation, or copyright infringement.

Warm Up

Answer the following questions.

(1) Choose the one in which availability is compromised in terms of information security from the options **A** to **C** below.

A A cyberattack caused a website to go down.

B Incorrect data was entered due to a keyboard typing error.

C Personal information was leaked due to a malware infection on the computer.

(2) Choose all actions that constitute a violation of the Unauthorized Computer Access Law from the options **A** to **D** below.

A Illegally used another person's user ID and password to access a computer.

B Stored a password that was obtained illegally on a computer.

C Shared a friend's user ID and password with someone else without the friend's permission.

D Published a website selling quasi-legal drugs or containing unlawful and inappropriate content.

Explanation

(1) Information security consists of three elements: integrity, confidentiality, and availability.

A When information becomes unavailable, availability is compromised.

B When information is no longer accurate, integrity is compromised.

C When unauthorized individuals can view the information, confidentiality is compromised.

Therefore, the correct answer is **A**.

(2) **A** Using a computer without having access rights constitutes unauthorized access and is prohibited under the Unauthorized Computer Access Law.

B Storing a password obtained illegally for the purpose of unauthorized access is prohibited under the Unauthorized Computer Access Law.

C Sharing someone else's password with a third party without valid reason or permission promotes unauthorized access and is also prohibited by law.

D Publishing illegal content may fall under network-related crimes.

Therefore, the correct answers are **A**, **B**, and **C**.

Try

1 There are three elements of information security: Integrity, Confidentiality, and Availability. For each of these elements, choose the one most appropriate measure from the following options **A** to **E**.

(1) Integrity (2) Confidentiality (3) Availability

A Report your arrival at work to the network administrator.

B Handle special data such as personal ID numbers in a room where only authorized personnel are allowed to enter.

C Keep logs of data access and modifications to enable traceability.

D Install backup power supplies for all devices related to critical information systems in preparation for power outages.

E Run malware in order to always have access to important information.

2 Choose the terms that best fit into the blanks [1] to [5] from the options **A** to **H** below, and answer using the letters.

When a third party without network access rights uses someone else's ID and password to illegally enter a computer system, it is called ([1]). Incidents have occurred in which individuals known as hackers or ([2]) have destroyed systems.

Programs that destroy internal computer data or cause abnormal operations are called ([3]). Among these are ([4]), which disguise themselves as legitimate programs and silently infiltrate systems to carry out attacks, and ([5]), which replicate themselves and spread across the internet like worms to increase infections.

A Worm **B** Cracker **C** Unauthorized access
D Trojan horse **E** Computer virus **F** Adware
G Keylogger **H** Impersonation

3 Answer the following questions.

(1) Choose all items from **A** to **D** that present a risk of computer virus infection.

A An email attachment sent from a computer infected with a virus.

B Connecting to a network infected with a virus.

C A USB memory stick that was used on a computer infected with a virus.

D A movie DVD played on a computer infected with a virus.

(2) Choose one act from **A** to **D** that constitutes a violation of the Unauthorized Computer Access Law.

A Provided personal information to a third party without the individual's consent.

B Took a photo of a magazine page with a smartphone and uploaded it to social media.

C Acquired a computer virus capable of automatically infiltrating networks.

D Used another person's user ID and password without permission to purchase products through online shopping.

Exercise

1 For each of the three elements of information security, select:

- The most appropriate description from Group A (**A**–**C**), one per element
- All applicable potential damages from Group B (a–f) that may occur if that element is not ensured

(1) Confidentiality (2) Integrity (3) Availability

<Group A>

A Ensuring uninterrupted access to information when needed.

B Ensuring that only authorized individuals can access the information.

C Ensuring that information has not been destroyed, tampered with, or deleted.

<Group B>

a Eavesdropping on the network

b Service interruptions such as system downtime

c Password leakage

d Tampering with or destruction of information

e Information leakage

f Unauthorized use of computers or networks

2 Answer the following questions.

(1) Choose the most suitable option for the blank from the options **A** to **G** below, and answer using the letters.

[1] A virus that displays advertisements the user did not intend to see is called ( ).

[2] The act of illegally infiltrating a computer to tamper with, erase, or steal data is called ( ).

[3] Software that leaks data stored on a computer to the outside is called ( ).

[4] Malicious software including computer viruses and Trojan horses is collectively referred to as ( ).

A Malware **B** Keylogger **C** Adware **D** Cracking

E Hacking **F** Phishing **G** Spyware

(2) Choose the one correct statement from **A** to **D** regarding computer virus infection.

A As long as the computer is connected to a network, there is always a risk of infection by a computer virus.

B If you do not connect to a computer or network and only transfer data using a USB memory stick, infection will not occur.

C If you avoid accessing harmful or illegal websites, you will not get infected.

D As long as you do not open emails, you are safe from infection, so being cautious with emails is sufficient.